Applied Econometrics: Session 3

PhD Intensive Course

This note reviews the statistical layer that sits on top of causal design. The main theme is the "most dangerous equation": the standard error of an average falls at rate $1/\sqrt{n}$, not at rate $1/n$. This simple fact explains why small groups often look extreme in rankings, why randomized experiments can be correct in design but noisy in finite samples, and why applied econometrics must report uncertainty together with point estimates. We cover estimands, estimators, estimates, sampling variation, standard errors, confidence intervals, hypothesis tests, p-values, power, multiple testing, and the relationship between uncertainty and causal interpretation.

Reading and objective. Read Chapter 03, "Stats Review: The Most Dangerous Equation," from *Causal Inference for the Brave and True*. The objective is not to memorize formulas mechanically. The objective is to learn how to read an empirical result as one noisy realization from a sampling process, and to separate three questions: what is the causal target, what is the estimator, and how uncertain is the estimate?

Why This Lecture Matters

The first two lectures were about causal meaning. We asked what it means to say that a treatment caused an outcome, how potential outcomes define causal effects, why selection bias is dangerous, and why regression is best understood as a conditional expectation approximation unless stronger assumptions give it causal content.

This lecture adds the statistical layer. Even when the causal design is excellent, the estimate in a paper is still computed from a finite sample. A different random sample, or a different random assignment, would usually have produced a different number. That movement from sample to sample is called sampling variation.

Key warning. A credible research design can produce an imprecise estimate. A precise estimate can still be biased. Identification and inference solve different problems.

An applied empirical claim should therefore contain at least four parts:

1. the target parameter or estimand;
2. the estimator used to learn about that target;

3. the numerical estimate obtained in the sample;
4. a measure of uncertainty, usually a standard error or confidence interval.

Estimand, Estimator, and Estimate

Definition 1 (Estimand)
An estimand is the population object we want to learn. In causal work it is often an average treatment effect,

$$\text{ATE} = \mathbb{E}[Y_1 - Y_0],$$

or an average treatment effect for a subpopulation, such as ATT.

Definition 2 (Estimator)
An estimator is a rule that maps data into a number. For example, in a randomized experiment with a treated group and a control group, a natural estimator of the average treatment effect is

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0.$$

Before the data are realized, $\hat{\tau}$ is a random variable.

Definition 3 (Estimate)
An estimate is the actual numerical value produced by applying an estimator to one dataset. If the treated mean is 76.3 *and the control mean is* 72.1*, the estimate is*

$$\hat{\tau} = 76.3 - 72.1 = 4.2.$$

Students often mix these objects. The estimand is the thing in the world we would like to know. The estimator is the recipe. The estimate is the number on the page. Standard errors describe the behavior of the recipe across repeated samples, not the moral importance of the number and not the truth of the identifying assumption.

Example 1 (Tutoring program)
Suppose a university randomly offers tutoring to half of first-year students. The causal estimand might be the average effect of tutoring on exam scores for this population. The estimator might be the treated mean minus the control mean. The estimate might be 4.2 *exam points. The standard error then tells us how much this difference-in-means estimator would typically move if the experiment were repeated.*

Sampling Variation

Imagine a large population of students with average exam score $\mu = 70$ and individual standard deviation $\sigma = 12$. If we draw one

classroom of 25 students, the classroom average might be 66. If we draw another classroom of 25 students, the average might be 73. Neither classroom average is the population mean. Each is one noisy estimate of the population mean.

Definition 4 (Sampling variation)
Sampling variation is the variation in an estimator across repeated samples generated by the same sampling process.

Definition 5 (Sampling distribution)
The sampling distribution of an estimator is the probability distribution of the estimator across repeated samples. The population distribution describes individual observations. The sampling distribution describes statistics computed from samples.

This distinction matters. A standard deviation describes the dispersion of individual outcomes. A standard error describes the dispersion of an estimator. For instance, students may differ a lot in exam scores, but if we observe a very large number of students, the average exam score may still be estimated very precisely.

The School-Ranking Trap

The Python Causality Handbook chapter begins with a simple but powerful idea: small groups are more likely to produce extreme averages. This is the logic behind Howard Wainer's phrase "the most dangerous equation."

Consider school rankings based on average test scores. Some schools have only 20 tested students, while others have 500. Even if all schools were equally good, the small schools would have much noisier average scores because the average is based on fewer observations. Therefore small schools would be overrepresented among the best observed schools. They would also be overrepresented among the worst observed schools.

> **Diagnostic for rankings.** If a small group appears surprisingly often at the top of a ranking, check the bottom of the ranking. If the same kind of small group also appears at the bottom, the pattern is probably about noise, not excellence.

Example 2 (Schools with no size effect)
Suppose school quality is independent of enrollment. Let the true school mean be

$$\mu_j = 70 + q_j, \qquad q_j \sim \mathcal{N}(0, 2^2),$$

and let the observed school average be

$$\bar{Y}_j = \mu_j + u_j, \qquad u_j \sim \mathcal{N}\left(0, \frac{12^2}{n_j}\right).$$

The variance of the noise term is larger when n_j is smaller. Thus the smallest schools naturally generate the most volatile observed averages.

The course script `lectures/code/lecture-3-stats-review.py` simulates this exact logic. It creates many schools with different enrollments, assigns true school quality independently of size, and then adds sampling noise with standard deviation $12/\sqrt{n_j}$. The script prints the average school size among top, middle, and bottom observed performers. The intended lesson is that extreme observed performance need not reveal extreme true quality.

The Most Dangerous Equation

Theorem 1 (Standard error of the sample mean)
Let $Y_1, \ldots, Y_n$ be independent observations with

$$\mathbb{E}[Y_i] = \mu, \qquad \mathrm{Var}(Y_i) = \sigma^2.$$

The sample mean

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$$

has variance

$$\mathrm{Var}(\bar{Y}) = \frac{\sigma^2}{n}$$

and standard error

$$\mathrm{SE}(\bar{Y}) = \frac{\sigma}{\sqrt{n}}.$$

Proof 1
Using independence,

$$\mathrm{Var}(\bar{Y}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right) = \frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^{n} Y_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(Y_i).$$

Since each variance is σ^2,

$$\mathrm{Var}(\bar{Y}) = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}.$$

Taking the square root gives $\mathrm{SE}(\bar{Y}) = \sigma/\sqrt{n}$.

The dangerous part is the square root. Precision improves slowly. To cut the standard error in half, we need four times as many observations.

Sample size n	$\mathrm{SE}(\bar{Y}) = 12/\sqrt{n}$	Interpretation
25	2.40	noisy average
100	1.20	half as noisy
400	0.60	half again
1600	0.30	half again

Common mistake. Do not say that increasing the sample from 100 to 200 cuts the standard error in half. It cuts the standard error by a factor of $\sqrt{100/200} = 1/\sqrt{2}$, about 0.71.

Standard Deviation Versus Standard Error

Standard deviation	Standard error
Spread of individual outcomes	Spread of an estimator
Describes students, workers, firms, schools	Describes $\bar{Y}$, $\hat{\beta}$, $\widehat{\mathrm{ATE}}$
Usually does not vanish with n	Usually shrinks as sample size grows
Answers: how different are units?	Answers: how precise is the estimate?

If students differ greatly in ability, the standard deviation of scores may be large. But if we observe thousands of students, the average score may be known quite precisely. Conversely, a small pilot study may have an average that moves a lot across samples even when individual outcomes are not especially variable.

Estimators Are Random Variables

Before data are observed, the sample mean is random:

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i.$$

After data are observed, $\bar{Y}$ becomes a number. In inference, we reason about the estimator before the data were realized. This is why we can say that a reported estimate is one point from a cloud of estimates that could have been obtained under repeated sampling.

Assumption 1 (Independent sampling)
For the elementary formulas in this lecture, observations are treated as independent draws from a common population distribution. Later in the

course, we will relax this with heteroskedasticity-robust, clustered, and time-series standard errors.

> **Mental model.** The estimate in a table is not the whole cloud. It is one draw from the cloud. The standard error estimates the cloud's width.

Bias and Variance

It is useful to separate two components of estimation error:

$$\hat{\theta} - \theta = \underbrace{\mathbb{E}[\hat{\theta}] - \theta}_{\text{bias}} + \underbrace{\hat{\theta} - \mathbb{E}[\hat{\theta}]}_{\text{sampling noise}} .$$

Identification assumptions aim at the first term. Standard errors describe the second term.

> **Bad inference habit.** A small standard error does not rescue a biased research design. It only says that the biased estimator is tightly estimated.

For causal work this distinction is essential. A randomized experiment can make $\mathbb{E}[\hat{\tau}]$ equal to the causal estimand, but in a small experiment $\hat{\tau}$ can still be far from the truth. An observational regression with omitted variables can have a tiny standard error and still estimate the wrong object.

Standard Error of a Mean

In practice, the population standard deviation σ is unknown. We estimate it with the sample standard deviation

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2}.$$

Then the estimated standard error of the mean is

$$\widehat{\mathrm{SE}}(\bar{Y}) = \frac{s}{\sqrt{n}}.$$

The denominator $n-1$ appears because $\bar{Y}$ is estimated from the same data. Once the deviations $Y_i - \bar{Y}$ are formed, they must add to zero, so only $n-1$ deviations are free to vary independently.

Example 3 (Exam scores)
Suppose a class has $\bar{Y} = 72$, $s = 12$, and $n = 25$. Then

$$\widehat{\mathrm{SE}}(\bar{Y}) = \frac{12}{\sqrt{25}} = 2.40.$$

A rough 95 percent margin of error is

$$1.96 \times 2.40 = 4.70.$$

With $n = 100$ *and the same standard deviation, the standard error is* 1.20 *and the rough margin is* 2.35.

Treatment-Control Differences

In a randomized experiment with binary treatment D_i, define

$$\bar{Y}_1 = \frac{1}{n_1} \sum_{D_i=1} Y_i, \qquad \bar{Y}_0 = \frac{1}{n_0} \sum_{D_i=0} Y_i.$$

The difference-in-means estimator is

$$\widehat{\text{ATE}} = \bar{Y}_1 - \bar{Y}_0.$$

Random assignment gives the causal interpretation; the standard error gives the precision.

Theorem 2 (Standard error of a difference in means)
If treatment and control sample means are independent, then

$$\text{Var}(\bar{Y}_1 - \bar{Y}_0) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.$$

The usual plug-in standard error is

$$\widehat{\text{SE}}(\bar{Y}_1 - \bar{Y}_0) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}.$$

Proof 2
Use

$$\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B) - 2\text{Cov}(A, B).$$

When the two sample means are independent, the covariance term is zero. The variance of each group mean is the group variance divided by the group sample size. Substitution gives the result.

Example 4 (Balanced and unbalanced experiments)
Suppose $s_1 = s_0 = 12$ *and the total sample size is* 200. *If* $n_1 = n_0 = 100$,

$$\widehat{\text{SE}} = \sqrt{\frac{12^2}{100} + \frac{12^2}{100}} = 1.70.$$

If instead $n_1 = 20$ *and* $n_0 = 180$,

$$\widehat{\text{SE}} = \sqrt{\frac{12^2}{20} + \frac{12^2}{180}} = 2.83.$$

For equal variances and a fixed total sample size, balanced groups are most precise.

The Central Limit Theorem

The central limit theorem is the reason that normal approximations appear so often in applied econometrics.

Theorem 3 (Central limit theorem, informal)
If $Y_1, \ldots, Y_n$ are independent draws with mean μ and finite variance σ^2, then for large n,

$$\frac{\bar{Y} - \mu}{\sigma / \sqrt{n}} \approx \mathcal{N}(0, 1).$$

The CLT does not say that individual outcomes are normal. It says that averages are approximately normal in large samples. This is why we can use normal critical values for sample means and many regression estimators even when the raw data are skewed.

> **Regularity matters.** The CLT can work poorly with very small samples, extreme outliers, heavy tails, strong dependence, or badly behaved sampling schemes. The formula is a workhorse, not a license to ignore the data generating process.

Confidence Intervals

For an estimate $\hat{\theta}$ with estimated standard error $\widehat{\mathrm{SE}}(\hat{\theta})$, the usual large-sample 95 percent confidence interval is

$$\hat{\theta} \pm 1.96\, \widehat{\mathrm{SE}}(\hat{\theta}).$$

The center is the estimate. The radius is about two standard errors. Wider intervals mean less precision.

Example 5 (Tutoring effect)
Suppose a randomized tutoring program gives

$$\hat{\tau} = 4.0, \qquad \widehat{\mathrm{SE}}(\hat{\tau}) = 1.5.$$

The 95 percent confidence interval is

$$4.0 \pm 1.96(1.5) = [1.06, 6.94].$$

The data are compatible with a modest positive effect and a large positive effect. Under the usual 95 percent rule, zero is not in the interval.

Where the Formula Comes From

The CLT gives

$$Z = \frac{\hat{\theta} - \theta}{\mathrm{SE}(\hat{\theta})} \approx \mathcal{N}(0, 1).$$

For a standard normal variable,

$$\mathbb{P}(-1.96 \leq Z \leq 1.96) \approx 0.95.$$

Substitute the statistic:

$$\mathbb{P}\left(-1.96 \leq \frac{\hat{\theta} - \theta}{\mathrm{SE}(\hat{\theta})} \leq 1.96\right) \approx 0.95.$$

Rearranging the inequalities gives

$$\mathbb{P}\left(\hat{\theta} - 1.96\mathrm{SE}(\hat{\theta}) \leq \theta \leq \hat{\theta} + 1.96\mathrm{SE}(\hat{\theta})\right) \approx 0.95.$$

In practice we replace the unknown standard error with an estimated standard error.

What 95 Percent Confidence Means

Frequentist confidence is about the procedure. If we repeated the study many times and computed an interval each time using the same formula, about 95 percent of those intervals would cover the true parameter.

After seeing one interval, the parameter is fixed. The interval either covers it or it does not. Strictly speaking, it is not correct to say that there is a 95 percent probability that this exact interval contains the true parameter.

> **Good language.** "Using a procedure that has about 95 percent coverage in repeated samples, we obtain the interval $[1.06, 6.94]$."

Do Not Compare Two Separate Intervals

To test whether treatment and control means differ, use the standard error for the difference:

$$(\bar{Y}_1 - \bar{Y}_0) \pm 1.96\sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}.$$

Separate confidence intervals for $\bar{Y}_1$ and $\bar{Y}_0$ answer separate questions. Their visual overlap is not the correct test of the treatment effect. The uncertainty must be attached to the estimator being interpreted.

Hypothesis Tests

A hypothesis test starts by assuming a benchmark value is true:

$$H_0 : \theta = \theta_0.$$

Then it asks whether the observed estimate is surprising under that benchmark. For a large-sample normal test, the test statistic is

$$z = \frac{\hat{\theta} - \theta_0}{\widehat{\mathrm{SE}}(\hat{\theta})}.$$

The numerator is distance from the null. The denominator is typical sampling noise. A value of $z = 2$ means that the estimate is about two standard errors away from the null.

> **Two-sided 5 percent rule.** Reject H_0 at the 5 percent level if $|z| > 1.96$.

Example 6 (Testing a tutoring effect)
Let $\hat{\tau} = 4.0$, $\widehat{\mathrm{SE}} = 1.5$, and test $H_0 : \tau = 0$. Then

$$z = \frac{4.0 - 0}{1.5} = 2.67.$$

Since $2.67 > 1.96$, we reject zero at the 5 percent level in a two-sided test.

Do not confuse rejecting a null with proving a theory. The test is conditional on assumptions about sampling, standard errors, and the null. It is one part of empirical reasoning, not the whole argument.

P-values

Definition 6 (P-value)
The p-value is the probability, under the null hypothesis, of observing a test statistic at least as extreme as the one actually observed.

For a two-sided normal test,

$$p = 2[1 - \Phi(|z|)],$$

where Φ is the standard normal cumulative distribution function.

The most common mistake is to read a p-value as

$$\mathbb{P}(H_0 \mid \text{data}).$$

That is not what it is. A p-value is closer to

$$\mathbb{P}(\text{data as or more extreme} \mid H_0).$$

> **Plain English.** A small p-value says the estimate would be unusual if the null were true. It does not say the probability that the null is true.

One-Sided and Two-Sided Tests

A two-sided alternative is

$$H_1 : \theta \neq \theta_0.$$

It rejects for estimates far above or far below the null. A one-sided alternative is

$$H_1 : \theta > \theta_0 \quad \text{or} \quad H_1 : \theta < \theta_0.$$

One-sided tests should be chosen before looking at the data and should be justified by the research question. In applied work, two-sided tests are the default.

Statistical Significance and Economic Importance

Statistical significance is not economic importance. With a huge dataset, a tiny effect can be statistically significant. With a small dataset, a large effect can be statistically imprecise.

Result	Estimate	Standard error
Huge dataset	0.03	0.01
Small study	4.00	2.50

The first result gives $z = 3$, so it is statistically significant at common levels, but an effect of 0.03 may be economically irrelevant. The second result gives $z = 1.6$, so it is not significant at the 5 percent level, but an effect of 4 units may matter a lot. Applied interpretation must discuss magnitude, uncertainty, and design.

Failure to Reject Is Not Evidence of Zero

Suppose

$$\hat{\theta} = 4.0, \qquad \widehat{\mathrm{SE}} = 3.0.$$

Testing $H_0 : \theta = 0$ gives

$$z = \frac{4.0}{3.0} = 1.33.$$

The 95 percent confidence interval is

$$4.0 \pm 1.96(3.0) = [-1.88, 9.88].$$

We cannot reject zero at the 5 percent level. But the interval also contains large positive effects. A better interpretation is: the estimate is too imprecise to distinguish zero from meaningful positive effects.

Bad sentence. "The treatment had no effect because the result is not statistically significant."
Better sentence. "The study is too imprecise to rule out either zero or economically meaningful positive effects."

Power

Definition 7 (Power)
Power is the probability of rejecting the null hypothesis when a specific alternative is true.

Power depends on the true effect size, the standard error, the significance level, and the form of the test. For a two-sided test of $H_0 : \theta = 0$, the test rejects when

$$|\hat{\theta}/\mathrm{SE}(\hat{\theta})| > 1.96.$$

If the true effect is small relative to the standard error, the test will reject rarely. If the true effect is large relative to the standard error, the test will reject often.

Example 7 (Approximate power logic)
Suppose the true effect is $\tau = 4$ *and the standard error is approximately* 2.4. *Then the estimator is roughly*

$$\hat{\tau} \sim \mathcal{N}(4, 2.4^2).$$

For a two-sided 5 percent test of zero, rejection requires $|\hat{\tau}| > 1.96(2.4) = 4.70$. *Since the true mean of the sampling distribution is only* 4, *many samples will fail to reject even though the effect is truly positive. The study is underpowered.*

Low-power studies have two bad features. First, they often miss real effects. Second, when they do reject, the significant estimates are often exaggerated in magnitude because only unusually large draws cross the significance threshold.

Multiple Testing

If one test is run at the 5 percent level and the null is true, the probability of a false rejection is 5 percent. If many tests are run, the chance of at least one false rejection can become large.

If m independent true null hypotheses are tested at level α, the probability of at least one false positive is

$$1 - (1 - \alpha)^m.$$

With $\alpha = 0.05$ and $m = 20$,

$$1 - 0.95^{20} \approx 0.64.$$

Thus, if we search across many outcomes, subgroups, and specifications, some small p-values can appear by chance.

> **Practical discipline.** Pre-specify main outcomes when possible. Report the family of results, not only the most impressive one. Label exploratory analysis as exploratory. Consider corrections such as Bonferroni, Holm, or false-discovery-rate procedures when the research design involves many tests.

Uncertainty and Causal Estimates

Causal interpretation and statistical precision answer different questions.

Problem	Main question
Identification	Is the estimator centered on the causal estimand?
Inference	How dispersed is the estimator around that center?
Measurement	Are treatment and outcome recorded correctly?
External validity	Does this population answer the policy question?

In a randomized experiment,

$$D_i \perp\!\!\!\perp (Y_{1i}, Y_{0i})$$

solves the selection problem in expectation. But the finite-sample difference in means still varies across random samples and random assignments:

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0.$$

Randomization removes systematic selection bias; it does not remove luck.

In observational designs, the standard error formula must match the sampling process and dependence structure. Heteroskedasticity, clustering, serial correlation, generated regressors, attrition, and noncompliance can all change the appropriate inference method.

> **Course habit.** Never use the word "causal" because a p-value is small. Use the word "causal" because the identifying variation and assumptions justify a causal interpretation.

Using the Python Example

The script

`lectures/code/lecture-3-stats-review.py`

is a companion to this lecture. From the repository root, run

```
python lectures/code/lecture-3-stats-review.py
```

or, to save figures,

```
python lectures/code/lecture-3-stats-review.py -save-plots.
```

The script contains two demonstrations.

1. **School rankings.** School size has no causal effect in the data generating process, but small schools appear in both tails of observed rankings because their averages have larger standard errors.
2. **Repeated experiments.** The true treatment effect is fixed, but repeated randomized studies produce different estimates. As sample size grows, the sampling distribution tightens around the truth.

When reading the output, focus on three quantities: the point estimate, the standard error, and the confidence interval. Then ask whether the interval is narrow enough to answer the substantive question.

How to Read a Results Table

When reading any empirical results table, do not start with stars. Read in this order:

1. What is the estimand: ATE, ATT, regression slope, reduced form, or something else?
2. What identifying assumption gives the estimate causal meaning?
3. What are the units and magnitude of the point estimate?
4. What is the standard error or confidence interval?
5. Is the estimate precise enough to answer the economic question?

6. Are there multiple outcomes, subgroups, or specifications that change how we should read the p-values?

   A useful writing template is:

   We estimate that X changes Y by $\hat{\theta}$ units ($\widehat{\text{SE}} = s$), with a 95 percent confidence interval of $[L, U]$. The identifying variation comes from .... The estimate is economically large/small because .... The main threat to causal interpretation is ....

Worked Problems

Problem 1: Standard Error of a Mean

A researcher observes $n = 64$ workers. Their average weekly hours are $\bar{Y} = 38$ and the sample standard deviation is $s = 8$.

Solution. The estimated standard error is

$$\widehat{\text{SE}}(\bar{Y}) = \frac{8}{\sqrt{64}} = 1.$$

The approximate 95 percent confidence interval is

$$38 \pm 1.96(1) = [36.04, 39.96].$$

Interpretation: the average is estimated with a margin of about two hours.

Problem 2: Difference in Means

A randomized training program has $n_1 = 100$ treated workers and $n_0 = 100$ control workers. Mean monthly earnings are $\bar{Y}_1 = 2200$ and $\bar{Y}_0 = 2100$. The sample standard deviations are $s_1 = 500$ and $s_0 = 600$.

Solution. The estimated effect is

$$\hat{\tau} = 2200 - 2100 = 100.$$

The standard error is

$$\widehat{\text{SE}}(\hat{\tau}) = \sqrt{\frac{500^2}{100} + \frac{600^2}{100}} = \sqrt{2500 + 3600} = 78.10.$$

The 95 percent confidence interval is

$$100 \pm 1.96(78.10) = [-53.08, 253.08].$$

This experiment is compatible with a negative effect, zero, and a moderately large positive effect. The point estimate is positive, but precision is weak.

Problem 3: Hypothesis Test and P-value

Using Problem 2, test $H_0 : \tau = 0$ against a two-sided alternative.

Solution. The test statistic is

$$z = \frac{100}{78.10} = 1.28.$$

Since $1.28 < 1.96$, we do not reject zero at the 5 percent level. The two-sided p-value is

$$p = 2[1 - \Phi(1.28)] \approx 0.20.$$

This does not prove the effect is zero. It says that an estimate this far from zero is not very unusual if the null were true.

Problem 4: Multiple Testing

A paper tests 10 independent outcomes at the 5 percent level, and suppose all 10 null hypotheses are true. What is the probability of at least one false positive?

Solution.

$$1 - (1 - 0.05)^{10} = 1 - 0.95^{10} \approx 0.40.$$

Even with no true effects, there is about a 40 percent chance of at least one significant result. This is why selective reporting is dangerous.

Problem 5: Causal Meaning Versus Precision

An observational regression estimates that private tutoring raises scores by 5 points with standard error 0.5. Should we conclude that tutoring causes scores to rise by 5 points?

Solution. Not from precision alone. The estimate is precise: a 95 percent interval is roughly $[4.02, 5.98]$. But causal interpretation requires an identifying assumption. Students who receive tutoring may differ in family income, motivation, prior preparation, or school quality. The small standard error only says the regression coefficient is tightly estimated; it does not solve selection bias.

Checklist for Students

1. Can I state the estimand in words and in notation?
2. Can I distinguish the estimator from the estimate?
3. Do I know what sampling process makes the estimator random?
4. Did I report the standard error in the same units as the estimate?

5. Did I compute the confidence interval for the estimator I interpret?
6. Did I avoid comparing two separate confidence intervals when the relevant object is a difference?
7. Did I interpret a p-value as $\mathbb{P}(\text{data as extreme} \mid H_0)$, not as $\mathbb{P}(H_0 \mid \text{data})$?
8. Did I separate statistical significance from economic importance?
9. Did I avoid saying "no effect" when the result is merely imprecise?
10. Did I check whether many outcomes, subgroups, or specifications were searched?
11. Did I explain the identifying assumption separately from the standard error?

Practice Questions

1. A sample has $n = 25$, $\bar{Y} = 10$, and $s = 5$. Compute the standard error and a 95 percent confidence interval for the mean.
2. If the sample size rises from 50 to 200 and the individual standard deviation is unchanged, by what factor does the standard error change?
3. Explain why small schools may appear more often among both the best and worst schools in a ranking based on average test scores.
4. In a randomized experiment, $\bar{Y}_1 = 84$, $\bar{Y}_0 = 80$, $s_1 = s_0 = 12$, and $n_1 = n_0 = 36$. Compute the treatment effect estimate, standard error, 95 percent confidence interval, and z statistic for testing zero.
5. Why is "the p-value is 0.03, so there is a 3 percent probability that the null is true" incorrect?
6. Give an example of an estimate that is statistically significant but economically unimportant.
7. Give an example of an estimate that is economically important but not statistically significant.
8. What is power? Why do small studies often have low power?
9. A paper reports 30 subgroup regressions and highlights the two with $p < 0.05$. What concern should you raise?

10. Explain in two sentences why randomization solves a selection problem but does not eliminate sampling variation.